Math 526, F09 **Name:** Answer
Quiz 4 **SSN:** xxx-xx-

<u>**Instructions:**</u> **There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (12 points) Given

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 5 \\ 0 & 4 & 0 \end{bmatrix}.$$

Find $\mathbf{A}^{-1}$ using the Gaussian-Jordan Elimination.

Solution:

$$[A \,\vdots\, I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 2 & 4 & 5 & 0 & 1 & 0 \\ 0 & 4 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{l_{21}=2} \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & -2 & 1 & 0 \\ 0 & 4 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{l_{32}=2} \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & -2 & 1 & 0 \\ 0 & 0 & -2 & 4 & -2 & 1 \end{array}\right]$$

Gaussian

$$\xrightarrow{l_{23}=-\frac{1}{2}} \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -2 & 4 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{l_{13}=-1} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 5 & -2 & 1 \\ 0 & 2 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -2 & 4 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{l_{12}=2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & -2 & \frac{3}{4} \\ 0 & \boxed{2} & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \boxed{-2} & 4 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{\text{normalize}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & -2 & \frac{3}{4} \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & -2 & 1 & -\frac{1}{2} \end{array}\right] = [I \,\vdots\, A^{-1}]$$

Jordan

$$\Rightarrow A^{-1} = \begin{bmatrix} 5 & -2 & \frac{3}{4} \\ 0 & 0 & \frac{1}{4} \\ -2 & 1 & -\frac{1}{2} \end{bmatrix}$$

Problem 2. (8 points) Let $\mathbf{A} = \begin{bmatrix} 3 & 3 & 4 \\ 1 & -1 & 3 \\ -2 & 4 & -1 \end{bmatrix}$. Find the **LU** factorization of **A**.

Solution:

$$A = \begin{bmatrix} 3 & 3 & 4 \\ 1 & -1 & 3 \\ -2 & 4 & -1 \end{bmatrix} \xrightarrow{\ell_{21}=\frac{1}{3}} \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & 5/3 \\ -2 & 4 & -1 \end{bmatrix}$$

$$\xrightarrow{\ell_{31}=-\frac{2}{3}} \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & 5/3 \\ 0 & 6 & 5/3 \end{bmatrix}$$

$$\xrightarrow{\ell_{32}=-3} \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & 5/3 \\ 0 & 0 & 20/3 \end{bmatrix}$$

$\Rightarrow A = LU$ with

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ -\frac{2}{3} & -3 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & \frac{5}{3} \\ 0 & 0 & \frac{20}{3} \end{bmatrix}$$